



chain,” says Ravi Kant Malhan, director, smartcities and special projects, Schneider Electric India.

“I think SMEs and SMBs will play a crucial role in taking smart technology to the smartcities in India,” believes Venkatesh Swaminathan, country head, The Attachmate Group, India.

With ICT industry in India witnessing a new high, the service and manufacturing industry depends on the cloud platform for successful execution of services in smartcities. Platforms like machine-to-machine (M2M) and device-to-device (D2D) are getting more familiar and common in the Indian market.

“Technology plays an important role in improving the efficiency and capacity of city services. Technology solutions can be implemented in a small timeframe to bring quick improvement of the city services delivery,” opines Prabhakar.

“We are in a digital age, where intelligent, interconnected devices are generating huge amounts of data, which is changing the game. Assisted by digital technology and data analysis and prediction tools, infrastructure management will increase efficiency, reduce costs and improve sustainability,” says Keshab Panda, chief executive and member of the board, L&T Technology Services.

With the Internet of Things (IoT) model to be functional in the country, smartapplications is one of the brightest perspectives of smartcities. The IoT will get a great boost in these cities, which will be fully loaded with

smartsensors, applications, smartdevices and smart control systems.

Another model of smartcities that can affect Indian Infrastructure and construction industry is the retrofitting model. Tech companies are of the opinion that this model serves as a viable model for creating smartinfrastructure in India.

“Each country and demography is prioritising as per their current infrastructure and technology availability,” adds Panda.

As per industry experts, some of the new trends and solutions that are making the cities smarter are:

- Crowd monitoring
- Disaster management
- Traffic management and prediction
- Roof-mounted windmills
- Energy-storage solutions
- Net-zero Buildings
- Software-defined sensors

Challenges

Smartcities, which consist of smartgrids and smartdevices, have to deal with numerous issues.

These projects in India are faced with many hurdles and problems such as connectivity, infrastructure and unorganised data. One of the key challenges is that, smart technologies that are currently being used for developing these cities are very new and are not exactly tried and tested in India. Data on infrastructure and real estate in India is also not adequate and digitised, and this can slow the rate of development of these smartcities.

“Real estate data is one of the many components to enable a city to become smart. Hence, this needs to be evaluated on a case-to-case basis,” adds Prabhakar.

According to Panda, “Ownership of data and related issues with intellectual property (IP) have not been fully addressed and understood. There is a wide debate going on with respect to who is the real owner of

the data generated by end devices. Is it the end user, service provider, application provider or the government?”

These are not trivial issues, especially in light of the need to strike a balance between the need to protect individual privacy, state’s duty to provide security to its citizens and operator’s need for data to make meaningful decisions.

“One of the challenges in front of smartcities will be, basic infrastructure being very old, and existing records and documentation being totally unavailable or inadequate, at best. Technology upgrades, especially in brown-field smartcities, will require time-consuming surveys and multi-department coordination to establish base-line scenarios,” says Malhan.

Stated in a PHD Chamber and Cushman & Wakefield Research report, “Currently, total private investment in infrastructure is very little, and most of these were implemented through a public-private-partnership (PPP) models. So far, policies for foreign direct investment (FDI) have not been in favor of investors, owing to a poor structure of the PPP itself.”

Further, “Another fault was identified with the tendering process. Typically, operation and maintenance component is not incorporated for engineering, procurement and construction (EPC) infrastructure upgrades. Performance based maintenance contracts (PBMC) need to be adopted more widely.”

“The major issue that technology providers are faced with is the delay in taking forward both the 100 smartcities and Delhi-Mumbai Industrial Corridor (DMIC) programmes. The overall speed of execution is very slow, which is leading to delays in request for proposals (RFPs) for technology solutions and consulting,” opines Prabhakar.

India’s population is increasing rapidly and creating new cities is one of the priorities of the government. The main challenge the government is left with is how to develop old ex-